Addition and subtraction of algebraic fractions

Same denominator — add the numerators. Different denominators — build them the same first, then add.

LESSON 12–14 | 3 × 40 MIN | ALGEBRA 8, §4 | STAGE 9 · 2.5

LESSON CLOCK

5' | 12' | 8' | 13' | 2'

Warm-up	5 min
Explanation	12 min
Interactive	8 min
Practice	13 min
Homework	2 min

LEARNING OBJECTIVES

- Add and subtract fractions with the same denominator.
- Add and subtract fractions with different denominators using the LCD.
- Handle subtraction correctly — bracket the whole numerator being subtracted.
- Simplify the result and state the permissible values.

TEXTBOOK

National	§4, pp. 22–26
Cambridge	Learner's Book pp. 40–44
Workbook	Workbook 2.5

01 Explanation

Same denominator

$$\frac{A}{C} \pm \frac{B}{C} = \frac{A \pm B}{C}$$

Keep the denominator, combine the numerators, then **always try to cancel the result** — the answer is often much simpler than the working suggests.

$$\frac{x}{x+3} + \frac{3}{x+3} = \frac{x+3}{x+3} = 1$$

Different denominators

FOUR STEPS, EVERY TIME

1. Factorise the denominators and find the LCD.
2. Find each additional factor and build both fractions up.
3. Combine the numerators over the single denominator.

4. Simplify the numerator, cancel if you can, state the permissible values.

$$\frac{1}{x-1} + \frac{1}{x+1} = \frac{(x+1)+(x-1)}{(x-1)(x+1)} = \frac{2x}{x^2-1}$$

The bracket that everyone forgets

SUBTRACTION

When you subtract a fraction, the minus sign belongs to the **whole** numerator, so put it in a bracket before you open it.

$$\frac{2x}{x^2-4} - \frac{1}{x-2} = \frac{2x-(x+2)}{(x-2)(x+2)} = \frac{x-2}{(x-2)(x+2)} = \frac{1}{x+2}$$

Writing $2x - x + 2$ instead of $2x - (x + 2)$ changes the answer completely.

A whole expression is a fraction too

Any expression can be written over the denominator 1, so it can join the sum:

$$1 - \frac{1}{x+1} = \frac{(x+1)-1}{x+1} = \frac{x}{x+1}$$

02 Worked examples

EXAMPLE 1 ***Simplify*** $\dfrac{5}{x^2 - 9} + \dfrac{1}{x + 3}$

1. $x^2 - 9 = (x - 3)(x + 3)$

   Factorise.

2. $LCD = (x - 3)(x + 3)$

   The second denominator is one of its factors.

3. $$\frac{5}{(x - 3)(x + 3)} + \frac{x - 3}{(x - 3)(x + 3)}$$

   Additional factor $(x - 3)$ for the second fraction.

4. $$\frac{5 + x - 3}{(x - 3)(x + 3)} = \frac{x + 2}{x^2 - 9}$$

   Combine and tidy the numerator.

ANSWER

$$\frac{x + 2}{x^2 - 9}, \quad x \neq \pm 3$$

EXAMPLE 2 ***Simplify*** $\dfrac{1}{x-1} - \dfrac{2}{x^2-1}$

1. $x^2 - 1 = (x - 1)(x + 1)$

   Factorise.

2. $LCD = (x - 1)(x + 1)$

   The first denominator needs the factor (x + 1).

3. $\dfrac{x+1}{(x-1)(x+1)} - \dfrac{2}{(x-1)(x+1)}$

   Build the first fraction up.

4. $\dfrac{(x+1)-2}{(x-1)(x+1)} = \dfrac{x-1}{(x-1)(x+1)}$

   Combine — keep the bracket.

5. $\dfrac{1}{x+1}$

   Cancel (x − 1). Always look for this last step.

ANSWER

$\dfrac{1}{x+1}$, $x \neq \pm 1$

EXAMPLE 3 ***Simplify*** $\dfrac{1}{x(x+1)} + \dfrac{1}{(x+1)(x+2)}$

1. $LCD = x(x+1)(x+2)$

   (x + 1) is shared, so it is taken once.

2. $\dfrac{x+2}{x(x+1)(x+2)} + \dfrac{x}{x(x+1)(x+2)}$

   Additional factors (x + 2) and x.

3. $\dfrac{2x+2}{x(x+1)(x+2)}$

   Combine the numerators.

4. $\dfrac{2(x+1)}{x(x+1)(x+2)} = \dfrac{2}{x(x+2)}$

   Factorise the numerator and cancel (x + 1).

ANSWER

$\dfrac{2}{x(x+2)}, \ x \neq 0, -1, -2$

03 Interactive model

Reveal the four steps one at a time and ask which step the class thinks comes next.

Adding and subtracting, step by step

$$\frac{1}{x-1} + \frac{1}{x+1}$$

1 $LCD = (x-1)(x+1)$

Different factors, so multiply them.

2 *additional factors:* $(x+1)$ *and* $(x-1)$

LCD ÷ each denominator.

3 $$\frac{(x+1)+(x-1)}{(x-1)(x+1)}$$

Build both up and combine.

4 $$\frac{2x}{x^2-1}$$

Tidy the numerator. Nothing cancels here.

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Add	Qo‘shish	Сложение
Subtract	Ayirish	Вычитание
Like fractions	Bir xil maxrajli kasrlar	Дроби с одинаковыми знаменателями
Unlike fractions	Har xil maxrajli kasrlar	Дроби с разными знаменателями
Combine	Birlashtirish	Объединить
Numerator	Surat	Числитель
Whole expression	Butun ifoda	Целое выражение
Simplify the result	Natijani soddalashtirish	Упростить результат

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

$\frac{3}{x} + \frac{5}{x}$ equals:

- $\frac{8}{2x}$
- $\frac{8}{x}$
- $\frac{15}{x}$
- $\frac{8}{x^2}$

$\frac{2x}{x^2 - 4} - \frac{1}{x - 2}$ equals:

- $\frac{1}{x + 2}$
- $\frac{2x - 1}{x^2 - 4}$
- $\frac{x + 2}{x^2 - 4}$
- $\frac{3x + 2}{x^2 - 4}$

$\frac{1}{a-b} + \frac{1}{b-a}$ equals:

- $\frac{2}{a-b}$
- 0
- $\frac{2}{b-a}$
- it cannot be combined

$\frac{x}{x+3} + \frac{3}{x+3}$ equals:

- $\frac{x+3}{2x+6}$
- 1
- $\frac{3x}{x+3}$
- x

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $\frac{3}{x} + \frac{5}{x}$

ANSWER $\frac{8}{x}$

2. $\frac{7}{a} - \frac{2}{a}$

ANSWER $\frac{5}{a}$

3. $\frac{x}{4} + \frac{x}{4}$

ANSWER $\frac{x}{2}$

4. $\frac{2a}{b} + \frac{3a}{b}$

ANSWER $\frac{5a}{b}$

5. $\frac{1}{2} + \frac{1}{x}$

ANSWER $\frac{x + 2}{2x}$

6. $\frac{1}{x} - \frac{1}{y}$

ANSWER $\frac{y - x}{xy}$

7. $\frac{3}{x + 1} + \frac{2}{x + 1}$

ANSWER $\frac{5}{x + 1}$

Medium · the standard question

7 PROBLEMS

1. $$\frac{1}{x-1} + \frac{1}{x+1}$$

ANSWER $\frac{2x}{x^2-1}$

2. $$\frac{1}{x-2} - \frac{1}{x+2}$$

ANSWER $\frac{4}{x^2-4}$

3. $$\frac{x}{x+3} + \frac{3}{x+3}$$

ANSWER $1, x \neq -3$

4. $$\frac{2}{x} + \frac{3}{x^2}$$

ANSWER $\frac{2x+3}{x^2}$

5. $$\frac{a}{a-b} - \frac{b}{a-b}$$

ANSWER $1, a \neq b$

6. $$\frac{1}{a} + \frac{1}{b} + \frac{1}{c}$$

ANSWER $\frac{bc+ac+ab}{abc}$

7. $$\frac{5}{x^2-9} + \frac{1}{x+3}$$

ANSWER $\frac{x+2}{x^2-9}$

Hard · stretch and reason

7 PROBLEMS

1. $$\frac{1}{x^2 - x} + \frac{1}{x^2 + x}$$

ANSWER $\frac{2}{x^2 - 1}$

2. $$\frac{1}{x - 3} + \frac{1}{x + 3}$$

ANSWER $\frac{2x}{x^2 - 9}$

3. $$\frac{1}{a - b} + \frac{1}{b - a}$$

ANSWER $0, a \neq b$

4. $$\frac{2x}{x^2 - 4} - \frac{1}{x - 2}$$

ANSWER $\frac{1}{x + 2}$

5. $$1 - \frac{1}{x + 1}$$

ANSWER $\frac{x}{x + 1}$

6. $$\frac{1}{x - 1} - \frac{2}{x^2 - 1}$$

ANSWER $\frac{1}{x + 1}$

7. $$\frac{1}{x(x + 1)} + \frac{1}{(x + 1)(x + 2)}$$

ANSWER $\frac{2}{x(x + 2)}$

07 Homework

Homework — 6 problems

Algebra 8, §4, pp. 22–26. Cancel the answer wherever you can.

1. $\frac{4}{x} + \frac{9}{x}$
2. $\frac{1}{x + 2} + \frac{1}{x - 2}$
3. $\frac{3}{x - 4} - \frac{3}{x + 4}$
4. $\frac{x}{x - 5} - \frac{5}{x - 5}$
5. $\frac{1}{x} + \frac{1}{x^2} + \frac{1}{x^3}$
6. $2 - \frac{3}{x - 1}$